

Chapter-10 Trigonometric Equation & General Values

• Trigonometric Equation

↳ An equation which involves one or more trigonometric functions of unknown variables (angles) is called trigonometric Equation.

E.g: $\sin x = \frac{1}{2}$

• Solution of Trigonometric Equation

↳ The solution of Trigonometric Equation is value of variables that satisfy Trigonometric equation. A trigonometric equation has infinite solution.

• Principle Solution

↳ A solution that lies between 0 and 2π is called principle solution.

• General solution of Trigonometric equations:

① $\sin x = 0 \Rightarrow x = n\pi, n \in \mathbb{Z}$

Proof

Here,

$\sin x = 0$ which satisfies for all integral values of x .

$$\sin x = \sin 0, \sin(\pm\pi), \sin(\pm 2\pi), \dots$$

$$\therefore x = 0, \pm\pi, \pm 2\pi, \dots$$

$$\therefore x = n\pi, n \in \mathbb{Z}$$

Integer

(ii) $\cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

Here,

$\cos x = 0$ which satisfies for all odd multiples of $\pi/2$.

$$\cos x = \cos \frac{\pi}{2}, \cos\left(\pm \frac{3\pi}{2}\right), \cos\left(\pm \frac{5\pi}{2}\right), \dots$$

$$\therefore x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$$

$$\therefore x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

(iii) $\tan x = 0 \Rightarrow x = n\pi, n \in \mathbb{Z}$

Proof

Here,

$$\tan x = 0$$

$$\Rightarrow \frac{\sin x}{\cos x} = 0$$

$$\Rightarrow \sin x = 0$$

$$\Rightarrow x = n\pi, n \in \mathbb{Z}$$

$$\text{iv) } \sin x = k = \sin \theta \Rightarrow x = n\pi + (-1)^n \theta, n \in \mathbb{Z}$$

Proof

$$\sin x = \sin \theta$$

$$\Rightarrow \sin x - \sin \theta = 0$$

$$\Rightarrow 2 \cos\left(\frac{x+\theta}{2}\right) \cdot \sin\left(\frac{x-\theta}{2}\right) = 0$$

Either,

$$\cos \frac{x+\theta}{2} = 0$$

$$\sin \frac{x-\theta}{2} = 0$$

$$\frac{x+\theta}{2} = (2n+1) \frac{\pi}{2}$$

$$\frac{x-\theta}{2} = n\pi$$

$$\Rightarrow x+\theta = (2n+1)\pi$$

$$x-\theta = 2n\pi$$

$$\Rightarrow x = (2n+1)\pi - \theta$$

$$x = 2n\pi + \theta$$

$$\therefore x = n\pi + (-1)^n \theta, n \in \mathbb{Z}$$

$$\text{v) } \cos x = k = \cos \theta \Rightarrow x = 2n\pi \pm \theta$$

Proof

$$\cos x = \cos \theta$$

$$\Rightarrow \cos x - \cos \theta = 0$$

$$\Rightarrow 2 \sin\left(\frac{\theta-x}{2}\right) \cdot \sin\left(\frac{x+\theta}{2}\right) = 0$$

$$\Rightarrow -2 \sin\left(\frac{x-\theta}{2}\right) \cdot \sin\left(\frac{x+\theta}{2}\right) = 0$$

Either,

$$-2 \sin\left(\frac{x-\theta}{2}\right) = 0$$

$$\sin\left(\frac{x+\theta}{2}\right) = 0$$

$$\sin\left(\frac{x-\theta}{2}\right) = 0$$

$$\frac{x+\theta}{2} = n\pi$$

$$\frac{x - \theta}{2} = n\pi$$

$$x + \theta = 2n\pi$$

$$x = 2n\pi - \theta$$

$$x - \theta = 2n\pi$$

$$x = 2n\pi + \theta$$

$$\therefore x = 2n\pi \pm \theta$$

vi) $\tan x = k = \tan \theta \Rightarrow x = n\pi + \theta$

proof: $\tan x = \tan \theta$

$$\text{or, } \frac{\sin x}{\cos x} = \frac{\sin \theta}{\cos \theta}$$

$$\text{or, } \sin x \cdot \cos \theta = \cos x \cdot \sin \theta$$

$$\text{or, } \sin x \cdot \cos \theta - \cos x \cdot \sin \theta = 0$$

$$\text{or, } \sin(x - \theta) = 0$$

$$\text{or, } x - \theta = n\pi$$

$$\boxed{\text{i.e. } x = n\pi + \theta}$$

Note :-

$$\text{For } \sin^2 x = \sin^2 \theta$$

$$\cos^2 x = \cos^2 \theta$$

$$\tan^2 x = \tan^2 \theta$$

$$\Rightarrow \boxed{x = n\pi \pm \theta}$$

Exercise 10

1) Find the principal solution ($0 \leq x < 2\pi$) of

a) $2\cos x - 1 = 0$

$$\text{or, } \cos x = \frac{1}{2}$$

$$\cos x = \cos \frac{\pi}{3}, \cos \left(2\pi - \frac{\pi}{3} \right)$$

$$\cos x = \cos \frac{\pi}{3}, \cos \frac{5\pi}{3}$$

$$\therefore x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$b) \sqrt{3} \sec x = 2$$

$$\Rightarrow \sec x = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \frac{1}{\cos x} = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \cos x = \frac{\sqrt{3}}{2}$$

$$\cos x = \cos \frac{\pi}{6}, \cos \left(2\pi - \frac{\pi}{6} \right)$$

$$\cos x = \cos \frac{\pi}{6}, \cos \frac{11\pi}{6}$$

$$\therefore x = \frac{\pi}{6}, \frac{11\pi}{6} \text{ h.}$$

$$c) \tan x = -\frac{1}{\sqrt{3}}$$

$$\text{or, } \tan x = \tan \left(\frac{\pi}{6} - \frac{\pi}{6} \right), \tan \left(2\pi - \frac{\pi}{6} \right)$$

$$\tan x = \tan \frac{5\pi}{6}, \tan \frac{11\pi}{6}$$

$$\therefore x = \frac{5\pi}{6}, \frac{11\pi}{6} \text{ h.}$$

$$d) \sin x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin x = \sin \frac{\pi}{4}, \sin \left(\pi - \frac{\pi}{4} \right)$$

$$\sin x = \sin \frac{\pi}{4}, \sin \frac{3\pi}{4}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4} \text{ etc.}$$

$$e) \sin x = \frac{\sqrt{3}}{2} \text{ and } \cos x = \frac{1}{2}$$

$$\sin x = \sin \frac{\pi}{3}$$

$$\cos x = \cos \frac{\pi}{3}$$

$$\therefore x = \frac{\pi}{3}$$

$$x = \frac{\pi}{3}$$

$$\therefore x = \frac{\pi}{3} \text{ etc.}$$

$$f) \tan x = -\frac{1}{\sqrt{3}} \text{ and } \sec x = -\frac{2}{\sqrt{3}}$$

$$\Rightarrow \tan x = \tan \left(\pi - \frac{\pi}{6} \right), \tan \left(2\pi - \frac{\pi}{6} \right)$$

$$\tan x = \tan \frac{5\pi}{6}, \tan \frac{11\pi}{6}$$

$$\Rightarrow x = \frac{5\pi}{6}, \frac{11\pi}{6}$$

$$\Rightarrow \sec x = -\frac{2}{\sqrt{3}}$$

$$\cos x = -\frac{\sqrt{3}}{2}$$

$$\cos x = \cos\left(\frac{\pi}{6} - \frac{\pi}{6}\right), \cos\left(\frac{\pi}{6} + \frac{\pi}{6}\right)$$

$$\cos x = \cos \frac{5\pi}{6}, \cos \frac{7\pi}{6}$$

$$\Rightarrow x = \frac{5\pi}{6}, \frac{7\pi}{6}$$

$$\therefore x = \frac{5\pi}{6}$$

2) Find the general solution

$$a) \cos^2 x = \frac{1}{2}$$

$$\cos^2 x = \left(\sqrt{\frac{1}{2}}\right)^2$$

$$\Rightarrow \cos^2 x = \frac{1}{\sqrt{2}} \cos^2 \frac{\pi}{4}$$

The general solution is,

$$n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$$

~~$$b) \cos 3x = \frac{1}{\sqrt{2}}$$~~

$$d) \tan^2 x = \frac{1}{3}$$

$$\Rightarrow \tan^2 x = \left(\frac{1}{\sqrt{3}}\right)^2$$

$$\Rightarrow \tan^2 x = \tan^2 \frac{\pi}{6}$$

The general solution is $x =$

$$n\pi \pm \frac{\pi}{6}$$

$$e) \sin x = \frac{1}{2} \text{ and } \cos x = -\frac{\sqrt{3}}{2}$$

Since $\sin x$ is positive and $\cos x$ is negative, the value lies in second quadrant.

$$\sin x = \frac{1}{2}$$

$$\Rightarrow \sin x = \sin\left(\pi - \frac{\pi}{6}\right)$$

$$\Rightarrow \sin x = \sin \frac{5\pi}{6}$$

The general solution is $x = 2n\pi + \frac{5\pi}{6}$

3.

$$b) \tan^3 x - 3\tan x = 0$$

$$\tan x (\tan^2 x - 3) = 0$$

Either,

$$\tan x = 0$$

$$\Rightarrow x = n\pi + 0$$

$$\Rightarrow x = n\pi, n \in \mathbb{Z}$$

$$\tan^2 x - 3 = 0$$

$$\text{or, } \tan^2 x = 3$$

$$\text{or, } \tan^2 x = (\sqrt{3})^2$$

$$\tan^2 x = \tan^2 \frac{\pi}{3}$$

The general solution is $x = n\pi \pm \frac{\pi}{3}$

$$\therefore x = n\pi, n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

4.

$$a) 4 \cos^2 x + 6 \sin^2 x = 5$$

$$\Rightarrow 4 - 4 \sin^2 x + 6 \sin^2 x = 5$$

$$\Rightarrow 4 + 2 \sin^2 x = 5$$

$$\Rightarrow 2 \sin^2 x = 1$$

$$\Rightarrow \sin^2 x = \frac{1}{2}$$

$$\Rightarrow \sin^2 x = \left(\frac{1}{\sqrt{2}} \right)^2$$

$$\Rightarrow \sin^2 x = \sin^2 \frac{\pi}{4}$$

The general solution is $x = n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}$

5)

$$b) \tan x + \sec x = \sqrt{3}$$

$$\frac{\sin x}{\cos x} + \frac{1}{\cos x} = \sqrt{3}$$

$$\text{or, } \sin x + 1 = \sqrt{3} \cos x$$

$$\text{or, } \sqrt{3} \cos x - \sin x = 1$$

Dividing both side by $\sqrt{(-1)^2 + (\sqrt{3})^2} = 2$

Now,

$$\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \frac{1}{2}$$

$$\sin \frac{\pi}{3} \cos \frac{\pi}{6} \cdot \cos x - \sin \frac{\pi}{6} \cdot \sin x = \cos \frac{\pi}{3}$$

$$\cos \left(\frac{\pi}{6} + x \right) = \cos \frac{\pi}{3}$$

∴ The General solution is

$$x + \frac{\pi}{6} = 2n\pi + \frac{\pi}{3}$$

Taking +ve

Taking -ve

$$x = 2n\pi + \frac{\pi}{3} - \frac{\pi}{6}$$

$$x = 2n\pi - \frac{\pi}{3} - \frac{\pi}{6}$$

$$\therefore x = 2n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$$

$$\therefore x = 2n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$$

c) $\sin x + \sqrt{3} \cos x = \sqrt{2}$

Dividing both side by $\sqrt{1^2 + (\sqrt{3})^2} = 2$

Now,

$$\frac{1}{2} \sin x + \frac{\sqrt{3}}{2} \cos x = \frac{\sqrt{2}}{2}$$

$$\cos \frac{\pi}{3} \cdot \sin x + \sin \frac{\pi}{3} \cdot \cos x = \frac{1}{\sqrt{2}}$$

$$\sin \left(x + \frac{\pi}{3} \right) = \sin \frac{\pi}{4}$$

∴ The General Solution is

$$x + \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}$$

$$x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}, n \in \mathbb{Z}$$

a) $\sqrt{3} \sin x - \cos x = \sqrt{2}$

Dividing both side by $\sqrt{(\sqrt{3})^2 + (-1)^2} = 2$

$$\frac{\sqrt{3}}{2} \sin x - \frac{\cos x}{2} = \frac{\sqrt{2}}{2}$$

$$\cos \frac{\pi}{3} \cdot \sin x - \cos x \cdot \sin \frac{\pi}{3} = \frac{\sqrt{2}}{2}$$

$$\sin\left(x - \frac{\pi}{3}\right) = \sin \frac{\pi}{4}$$

The general solution is,

$$x - \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}$$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{3}, n \in \mathbb{Z}$$

d) $\sqrt{2} \sec x + \tan x = 1$

or, $\frac{\sqrt{2}}{\cos x} + \frac{\sin x}{\cos x} = 1$

or, $\sqrt{2} \sec x + \tan x = \cos x$

or, $\cos x - \sin x = \sqrt{2}$

dividing both side by $\sqrt{1^2 + (-1)^2} = \sqrt{2}$

$$\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x = \frac{\sqrt{2}}{\sqrt{2}}$$

or, $\cos \frac{\pi}{4} \cdot \cos x - \sin \frac{\pi}{4} \cdot \sin x = 1$

or, $\cos\left(x + \frac{\pi}{4}\right) = \cos 0$

The general solution is,

$$x + \frac{\pi}{4} = 2n\pi \pm 0$$

$$\therefore x = 2n\pi - \frac{\pi}{4}$$

$$c) \sin x + \cos x = -\frac{1}{\sqrt{2}}$$

→ dividing both side by $\sqrt{1^2+1^2} = \sqrt{2}$

$$\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = -\frac{1}{\sqrt{2}}$$

$$\text{or, } \sin \frac{\pi}{4} \cdot \sin x + \cos \frac{\pi}{4} \cdot \cos x = -\frac{1}{2}$$

$$\text{or, } \cos x \cdot \cos \frac{\pi}{4} + \sin x \cdot \sin \frac{\pi}{4} = -\frac{1}{2}$$

$$\text{or, } \cos \left(x - \frac{\pi}{4} \right) = \cos \frac{2\pi}{3}$$

The general Solution is

$$x - \frac{\pi}{4} = 2n\pi \pm \frac{2\pi}{3}$$

$$\therefore x = 2n\pi \pm \frac{2\pi}{3} + \frac{\pi}{4}$$

4.

$$b) \cos^2 x - \sin^2 x + \cos x = 0$$

$$\Rightarrow \cos^2 x - (1 - \cos^2 x) + \cos x = 0$$

$$\text{or, } \cos^2 x - 1 + \cos^2 x + \cos x = 0$$

$$\text{or, } 2\cos^2 x + \cos x - 1 = 0$$

$$\text{or, } 2\cos^2 x + 2\cos x - \cos x - 1 = 0$$

$$\text{or, } 2\cos x (\cos x + 1) - 1(\cos x + 1) = 0$$

$$\text{or, } (2\cos x - 1)(\cos x + 1) = 0$$

Either,

$$2\cos x - 1 = 0$$

$$\cos x = \frac{1}{2}$$

$$\cos x = \cos \frac{\pi}{3}$$

The general solution is

$$x = 2n\pi \pm \frac{\pi}{3}$$

$$\text{or, } \cos x + 1 = 0$$

$$\cos x = -1$$

$$\cos x = \cos \pi$$

The general solution is

$$x = 2n\pi \pm \pi$$

$$\therefore x = 2n\pi \pm \frac{\pi}{3}, 2n\pi \pm \pi$$

$$c) 3 \cos^2 x + 5 \sin^2 x = 4$$

$$\text{or, } 3 \cos^2 x + 5(1 - \cos^2 x) = 4$$

$$\text{or, } 3 \cos^2 x + 5 - 5 \cos^2 x = 4$$

$$\text{or, } 5 - 2 \cos^2 x = 4$$

$$\text{or, } 2 \cos^2 x = 1$$

$$\text{or, } \cos^2 x = \frac{1}{2}$$

$$\text{or, } \cos^2 x = \left(\frac{1}{\sqrt{2}} \right)^2$$

$$\text{or, } \cos^2 x = \cos^2 \frac{\pi}{4}$$

The general solution is

$$x = n\pi \pm \frac{\pi}{4}$$

$$4 \sin^2 x - 8 \cos x + 1 = 0$$

$$\text{or, } 4(1 - \cos^2 x) - 8 \cos x + 1 = 0$$

$$\text{or, } 4 - 4 \cos^2 x - 8 \cos x + 1 = 0$$

$$\text{or, } -4 \cos^2 x + 8 \cos x - 5 = 0$$

$$\text{or, } 4 \cos^2 x + 10 \cos x - 2 \cos x - 5 = 0$$

$$\text{or, } 2 \cos x (2 \cos x + 5) - 1(2 \cos x + 5) = 0$$

$$\text{or, } (2 \cos x - 1)(2 \cos x + 5) = 0$$

Either,

$$2 \cos x - 1 = 0$$

$$\cos x = \frac{1}{2} = \cos \frac{\pi}{3}$$

The general solution is

$$x = 2n\pi \pm \frac{\pi}{3}$$

Or,

$$2 \cos x + 5 = 0$$

$$2 \cos x = -5$$

$$\cos x = -\frac{5}{2} \text{ (Impossible)}$$

$$\cos x =$$

$$\Rightarrow 2 \tan x = \cot x = \frac{1}{\tan x}$$

$$\Rightarrow \frac{2 \sin x}{\cos x} = \frac{\cos x}{\sin x}$$

$$\Rightarrow 2 \sin^2 x = \cos^2 x = 1 - \sin^2 x$$

$$\Rightarrow 2 \sin^2 x + \sin^2 x - \cos^2 x = 0$$

$$\Rightarrow 3 \sin^2 x + \sin x \cos x - 1 + \sin^2 x = 0$$

$$\Rightarrow 4 \sin^2 x + \sin x \cos x = 1$$

$$\Rightarrow 4 \sin x (3 \sin x + \cos x) = 1$$

$$e, 2 \tan x - \cot x = -1$$

$$\Rightarrow 2 \tan x - \frac{1}{\tan x} = -1$$

$$\Rightarrow 2 \tan^2 x - 1 = -\tan x$$

$$\Rightarrow 2 \tan^2 x + \tan x - 1 = 0$$

$$\Rightarrow 2 \tan^2 x + 2 \tan x - \tan x - 1 = 0$$

$$\Rightarrow 2 \tan x (\tan x + 1) - 1 (\tan x + 1) = 0$$

$$\Rightarrow (2 \tan x - 1) (\tan x + 1) = 0$$

Either,

$$2 \tan x - 1 = 0$$

$$\tan x = \frac{1}{2}$$

$$\tan x = \tan \tan^{-1} \frac{1}{2}$$

$$\therefore x = n\pi + \tan^{-1} \frac{1}{2}$$

$$\tan x + 1 = 0$$

$$\tan x = -1$$

$$\tan x = \tan \frac{3\pi}{4}$$

General soln is

$$x = n\pi + \frac{3\pi}{4}$$

$$7. b) 2\sin^2 x + \sin^2 2x = 2$$

$$\text{or, } 2\sin^2 x + 4\sin^2 x \cdot \cos^2 x = 2$$

$$\text{or, } \sin^2 x + 2\sin^2 x \cdot \cos^2 x = 1$$

$$\text{or, } \sin^2 x + 2\sin^2 x (1 - \sin^2 x) = 1$$

$$\text{or, } \sin^2 x + 2\sin^2 x - 2\sin^4 x = 1$$

$$\text{or, } 3\sin^2 x - 2\sin^4 x = 1$$

$$\text{or, } 2\sin^4 x - 3\sin^2 x + 1 = 0$$

$$\text{or, } 2\sin^4 x - 2\sin^2 x - \sin^2 x + 1 = 0$$

$$\text{or, } 2\sin^2 x (\sin^2 x - 1) - 1(\sin^2 x - 1) = 0$$

$$\text{or, } (2\sin^2 x - 1)(\sin^2 x - 1) = 0$$

Either,

$$2\sin^2 x - 1 = 0$$

$$\sin^2 x = \frac{1}{2}$$

$$\sin^2 x = \left(\frac{1}{\sqrt{2}}\right)^2$$

$$\sin^2 x = \sin^2 \frac{\pi}{4}$$

general solution is $n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$

or,

$$\sin^2 x - 1 = 0$$

$$\sin^2 x = 1$$

$$\sin^2 x = \sin^2 \frac{\pi}{2}$$

general solution is $n\pi + \frac{\pi}{2}$, $n \in \mathbb{Z}$

$$\therefore x = n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{2}, n \in \mathbb{Z} \quad \text{Ans.}$$

$$2b) \cos 3x = -\frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos 3x = \cos \frac{3\pi}{4}$$

$\therefore$ general solution is

$$3x = 2n\pi \pm \frac{3\pi}{4}$$

$$\text{or, } x = \frac{2n\pi \pm \frac{3\pi}{4}}{3}$$

$$\therefore x = \frac{2n\pi}{3} \pm \frac{\pi}{4}, n \in \mathbb{Z}$$

$$c) \cos 3x = \sin 2x$$

$$\Rightarrow \cos 3x = \cos \left(\frac{\pi}{2} - 2x \right)$$

The general solution is

$$3x = 2n\pi \pm \frac{\pi}{2} - 2x$$

$$\therefore x = \frac{2n\pi \pm \frac{\pi}{2} - \frac{2}{3}x}{3}, n \in \mathbb{Z}$$

$$f) \sec x = -2 \text{ and } \cot x = \frac{1}{\sqrt{3}}$$

~~The~~ Since, the value of $\sec$ is negative and $\cot$ is positive so it lies in 3rd quadrant.

Now,

$$\sec x = -2$$

$$\frac{1}{\cos x} = -2$$

$$\cos x = -\frac{1}{2}$$

$$\cos x = \cos \frac{4\pi}{3}$$

∴ General solution is

$$x = 2n\pi \pm \frac{4\pi}{3}$$

3.)

$$a) \sin 2x + \cos x = 0$$

$$\Rightarrow \sin 2x + \sin(\pi - x) = 0$$

$$\Rightarrow 2 \sin\left(\frac{2x + \pi - x}{2}\right) \cdot \cos\left(\frac{2x - \pi + x}{2}\right) = 0$$

$$\Rightarrow \sin\left(\frac{3\pi + x}{2}\right) \cdot \cos\left(\frac{\pi + x}{2}\right) = 0$$

Either,

$$\sin\left(\frac{3\pi + x}{2}\right) = 0$$

$$\sin\left(\frac{3\pi + x}{2}\right) = \sin 0$$

General solution is,

$$\frac{3\pi + x}{2} = n\pi$$

$$3\pi + x = 2n\pi$$

$$x = 2n\pi - 3\pi$$

$$\text{or, } \cos\left(\frac{\pi + x}{2}\right) = 0$$

~~cos~~ General solution is,

$$\frac{\pi + x}{2} = (2n + 1) \frac{\pi}{2}$$

~~$$x = (2n+1)\pi - \pi$$~~

$$c) \sin a n + \cos b n = 0$$

$$\text{or, } \cos b n = -\sin a n$$

$$\text{or, } \cos b n = \cos \left(\frac{\pi}{2} + a n \right)$$

The general solution is

$$b n = 2n\pi \pm \left(\frac{\pi}{2} + a n \right)$$

Taking +ve

$$b n = 2n\pi + \frac{\pi}{2} + a n$$

$$n = \frac{2n\pi}{b} + \frac{\pi}{2b} + \frac{a n}{b}, n \in \mathbb{Z}$$

Taking -ve

$$b n = 2n\pi - \frac{\pi}{2} - a n$$

$$n = \frac{2n\pi}{b} - \frac{\pi}{2b} - \frac{a n}{b}, n \in \mathbb{Z}$$

$$\therefore n = \frac{2n\pi}{b} + \frac{\pi}{2b} + \frac{a n}{b}, \frac{2n\pi}{b} - \frac{\pi}{2b} - \frac{a n}{b}, n \in \mathbb{Z}$$

$$Q) \sin 2n + \cos n = 0$$

$$\text{or, } \sin 2n + \sin \left(\frac{\pi}{2} - n \right) = 0$$

$$\text{or, } 2 \sin \left(\frac{2n + \pi/2 - n}{2} \right) \cdot \cos \left(\frac{2n - \pi/2 + n}{2} \right) = 0$$

$$\text{or, } \sin \left(\frac{2n + \pi/2 - n}{2} \right) \cdot \cos \left(\frac{2n - \pi/2 + n}{2} \right) = 0$$

$$\text{or, } \sin\left(\frac{\pi}{2} - \frac{\pi}{4}\right) \cdot \cos\left(\frac{3\pi}{2} - \frac{\pi}{4}\right) = 0$$

Either,

$$\sin\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = 0$$

general solution is

$$\frac{\pi}{2} - \frac{\pi}{4} = n\pi$$

$$\frac{\pi}{2} = n\pi + \frac{\pi}{4}$$

$$n = 2n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$$

$$\cos\left(\frac{3\pi}{2} - \frac{\pi}{4}\right) = 0$$

general solution is

$$\frac{3\pi}{2} - \frac{\pi}{4} = (2n+1)\frac{\pi}{2}$$

$$\frac{3\pi}{2} = (2n+1)\frac{\pi}{2} + \frac{\pi}{4}$$

$$n = (2n+1)\frac{\pi}{2} \times \frac{2}{3} + \frac{\pi}{2} \times \frac{2}{3}$$

$$n = (2n+1)\frac{\pi}{3} + \frac{\pi}{6}, n \in \mathbb{Z}$$

$$\therefore n = 2n\pi + \frac{\pi}{2}, (2n+1)\frac{\pi}{3} + \frac{\pi}{6}, n \in \mathbb{Z}$$

d) $\tan x + \cot x = 2$

$\Rightarrow \tan x + \frac{1}{\tan x} = 2$

$\Rightarrow \tan^2 x + 1 = 2 \tan x$

$\Rightarrow \tan^2 x - 2 \tan x + 1 = 0$

$\Rightarrow \tan^2 x - \tan x - \tan x + 1 = 0$

$\Rightarrow \tan x (\tan x - 1) - 1 (\tan x - 1) = 0$

$\Rightarrow (\tan x - 1)^2 = 0$

$\tan x - 1 = 0$

$\tan x = 1 \Rightarrow \tan \pi/4$

The general solution is,

$x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$

6. Solve

a) $\cos x + \cos 2x + \cos 3x = 0$

or, $\cos x + \cos 3x + \cos 2x = 0$

or, $2 \cos\left(\frac{x+3x}{2}\right) \cdot \cos\left(\frac{3x-x}{2}\right) + \cos 2x = 0$

or, $2 \cos 2x \cdot \cos x + \cos 2x = 0$

or, $\cos 2x (2 \cos x + 1) = 0$

Either,

$\cos 2x = 0$

The general soln is

$2x = (2n+1) \frac{\pi}{2}$

or, $x = (2n+1) \frac{\pi}{4}, n \in \mathbb{Z}$

Or,

$2 \cos x + 1 = 0$

$\cos x = -\frac{1}{2}$

~~The~~ $\cos x = \cos \frac{2\pi}{3}$

The general soln is

$x = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{Z}$

$$b) \sin 3x + \sin x = \sin 2x$$

$$\text{or, } 2 \sin\left(\frac{2x+x}{2}\right) \cdot \cos\left(\frac{3x-x}{2}\right) = \sin 2x$$

$$\text{or, } 2 \sin 2x \cdot \cos x - \sin 2x = 0$$

$$\text{or, } \sin 2x (2 \cos x - 1) = 0$$

Either,

$$\sin 2x = 0$$

or,

$$2 \cos x - 1 = 0$$

$$\text{The general soln is } \cos x = \frac{1}{2}$$

$$2x = n\pi$$

$$x = \frac{n\pi}{2}, n \in \mathbb{Z} \quad *$$

$$\cos x = \cos \frac{\pi}{3}$$

The general soln is

$$x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$c) \sin 2x + \sin 4x + \sin 6x = 0$$

$$\text{or } \sin 2x + \sin 6x + \sin 4x = 0$$

$$\text{or, } 2 \sin\left(\frac{6x+2x}{2}\right) \cdot \cos\left(\frac{6x-2x}{2}\right) + \sin 4x = 0$$

$$\text{or, } 2 \sin 4x \cdot \cos 2x + \sin 4x = 0$$

$$\text{or, } \sin 4x (2 \cos 2x + 1) = 0$$

Either,

$$\sin 4x = 0$$

or,

$$2 \cos 2x + 1 = 0$$

$$\text{The general soln is } \cos 2x = -\frac{1}{2}$$

$$4x = n\pi$$

$$x = \frac{n\pi}{4}, n \in \mathbb{Z} \quad *$$

$$\cos 2x = \cos \frac{2\pi}{3}$$

The general soln is

$$2x = 2n\pi \pm \frac{2\pi}{3}$$

$$\therefore x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z} \quad *$$

$$d) \sin x + \sin 3x + \sin 5x = 0$$

$$or, \sin x + \sin 5x + \sin 3x = 0$$

$$or, 2 \sin \left(\frac{5x+x}{2} \right) \cdot \cos \left(\frac{5x-x}{2} \right) + \sin 3x = 0$$

$$or, 2 \sin 3x \cdot \cos 2x + \sin 3x = 0$$

$$or, \sin 3x (2 \cos 2x + 1) = 0$$

Either,

or,

$$\sin 3x = 0$$

$$2 \cos 2x + 1 = 0$$

The general solⁿ is

$$\cos 2x = -\frac{1}{2}$$

$$3x = n\pi$$

$$x = \frac{n\pi}{3}, n \in \mathbb{Z}$$

$$\cos 2x = \cos \frac{2\pi}{3}$$

The general soln is,

$$2x = \cancel{2n\pi} \pm 2n\pi \pm \frac{2\pi}{3}$$

$$x = n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$e) \cos 3x + \cos x - \cos 2x = 0$$

$$\Rightarrow 2 \cos \left(\frac{3x+x}{2} \right) \cdot \cos \left(\frac{3x-x}{2} \right) - \cos 2x = 0$$

$$\Rightarrow 2 \cos 2x \cdot \cos x - \cos 2x = 0$$

$$\Rightarrow \cos 2x (2 \cos x - 1) = 0$$

Either,

or,

$$\cos 2x = 0$$

$$2 \cos x - 1 = 0$$

The general solⁿ is

$$\cos x = \frac{1}{2}$$

$$\cancel{2x} = (2n+1) \frac{\pi}{2}$$

$$\cos x = \cos \frac{\pi}{3}$$

$$\therefore x = (2n+1) \frac{\pi}{4}, n \in \mathbb{Z}$$

The general solⁿ is

$$x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$f) \cos x + \sin x = \cos 2x + \sin 2x$$

$$\Rightarrow \cos 2x - \cos x = \sin x - \sin 2x$$

$$\Rightarrow \cancel{\neq} \sin\left(\frac{2x+x}{2}\right) \cdot \sin\left(\frac{2x-x}{2}\right) = \cancel{\neq} \sin\left(\frac{2x-x}{2}\right) \cdot \cos\left(\frac{2x+x}{2}\right)$$

$$\Rightarrow \sin\left(\frac{3x}{2}\right) = \cos\left(\frac{3x}{2}\right)$$

$$\Rightarrow \cos\left(\frac{\pi}{2} - \frac{3x}{2}\right) = \cos\left(\frac{3x}{2}\right)$$

The general formula/solution is,

$$\cancel{x} \quad \frac{3x}{2} = 2n\pi \pm \left(\frac{\pi}{2} - \frac{3x}{2}\right)$$

$$\cancel{x} \quad \left[2n\pi + \left(\frac{\pi}{2} - \frac{3x}{2}\right)\right] \times \frac{2}{3}$$

$$\cancel{\therefore x = \frac{4n\pi}{3} \pm \left(\frac{\pi}{3} - x\right)} \quad \cancel{x \in \mathbb{Z}}$$

Taking +ve

$$\frac{3x}{2} + \frac{3x}{2} = 2n\pi + \frac{\pi}{2}$$

$$\therefore x = \frac{2n\pi}{3} + \frac{\pi}{6}, \quad n \in \mathbb{Z}$$

Taking -ve,

$$\frac{3x}{2} = 2n\pi - \frac{\pi}{2} + \frac{3x}{2}$$

$$g) \sin 2x \cdot \tan x + 1 = \sin 2x + \tan x$$

$$\Rightarrow \sin 2x \cdot \tan x - \tan x = \sin 2x - 1$$

$$\Rightarrow \tan x (\sin 2x - 1) - 1 (\sin 2x - 1) = 0$$

$$\Rightarrow (\tan x - 1) (\sin 2x - 1) = 0$$

either,

or,

$$\tan x - 1 = 0$$

$$\sin 2x - 1 = 0$$

$$\tan x = 1$$

$$\sin 2x = 1$$

$$\tan x = \tan \frac{\pi}{4}$$

$$\sin 2x = \sin \frac{\pi}{2}$$

general solution is

general solution is

$$\therefore x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$$

$$2x = n\pi + (-1)^n \frac{\pi}{2}$$

$$\therefore x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}, n \in \mathbb{Z}$$

$$h) \tan x + \tan 2x = 1 - \tan x \cdot \tan 2x$$

$$\Rightarrow \frac{\tan 2x + \tan x}{1 - \tan 2x \cdot \tan x} = 1$$

$$\Rightarrow \tan (2x + x) = \tan \frac{\pi}{4}$$

$$\Rightarrow \tan 3x = \tan \frac{\pi}{4}$$

General solution is

$$3x = n\pi + \frac{\pi}{4}$$

$$\therefore x = \frac{n\pi}{3} + \frac{\pi}{12}, n \in \mathbb{Z}$$

$$i) \tan x + \tan 2x + \sqrt{3} \tan x \cdot \tan 2x = \sqrt{3}$$

$$\text{or, } \tan x + \sqrt{3} \tan x \cdot \tan 2x + \tan 2x - \sqrt{3} = 0$$

$$\text{or, } \tan x + \tan 2x = \sqrt{3} - \sqrt{3} \tan x \cdot \tan 2x$$

$$\text{or, } \tan x + \tan 2x = \sqrt{3} (1 - \tan x \cdot \tan 2x)$$

$$\text{or, } \frac{\tan x + \tan 2x}{1 - \tan x \cdot \tan 2x} = \sqrt{3}$$

$$\tan (x + 2x) = \sqrt{3}$$

$$\text{or } \tan(\pi + 2\pi) = \tan \frac{\pi}{3}$$

$$\text{or } \tan 3\pi = \tan \frac{\pi}{3}$$

The General solution is,

$$3\pi = n\pi + \frac{\pi}{3}$$

$$\therefore \pi = \frac{n\pi}{3} + \frac{\pi}{9}, n \in \mathbb{Z}$$

$$j) \tan 3\pi + \tan 4\pi + \tan 7\pi = \tan 3\pi \cdot \tan 4\pi \cdot \tan 7\pi$$

$$\Rightarrow \tan 3\pi + \tan 7\pi = \tan 3\pi \cdot \tan 4\pi \cdot \tan 7\pi - \tan 4\pi$$

$$\Rightarrow \tan 3\pi + \tan 7\pi = -\tan 4\pi (1 - \tan 3\pi \cdot \tan 7\pi)$$

$$\Rightarrow \frac{\tan 3\pi + \tan 7\pi}{1 - \tan 3\pi \cdot \tan 7\pi} = -\tan 4\pi$$

$$\Rightarrow \tan(3\pi + 7\pi) = -\tan 4\pi$$

$$\Rightarrow \tan 10\pi = -\tan 4\pi$$

General solution is,

$$10\pi = n\pi - 4\pi$$

$$10\pi + 4\pi = n\pi$$

$$14\pi = n\pi$$

$$\therefore \pi = \frac{n\pi}{14}, n \in \mathbb{Z}$$

7) Solve:

$$a) \tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) = 4$$

$$= \frac{\tan \pi/4 + \tan \theta}{1 - \tan \pi/4 \cdot \tan \theta} + \frac{\tan \pi/4 - \tan \theta}{1 + \tan \pi/4 \cdot \tan \theta} = 4$$

$$= \frac{1 + \tan \theta}{1 - \tan \theta} + \frac{1 - \tan \theta}{1 + \tan \theta} = 4$$

$$\Rightarrow \frac{(1 + \tan \theta)^2}{1 - \tan^2 \theta} + \frac{(1 - \tan \theta)^2}{1 + \tan^2 \theta} = 4$$

$$\Rightarrow \frac{1 + 2\tan \theta + \tan^2 \theta}{1 - \tan^2 \theta} + \frac{1 - 2\tan \theta + \tan^2 \theta}{1 + \tan^2 \theta} = 4$$

$$\Rightarrow \frac{2 + 2\tan^2 \theta}{1 - \tan^2 \theta} = 4$$

$$\Rightarrow 2(1 + \tan^2 \theta) = 4(1 - \tan^2 \theta)$$

$$\Rightarrow 1 + \tan^2 \theta = 2 - 2\tan^2 \theta$$

$$\Rightarrow 3\tan^2 \theta = 1$$

$$\Rightarrow \tan^2 \theta = \left(\frac{1}{\sqrt{3}}\right)^2 \Rightarrow \tan \frac{\pi}{6}$$

The general solution is

$$\theta = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$$

~~$$\tan 2x + \sin 2x = 2$$~~

~~$$\Rightarrow \tan p = \cot q$$~~

~~$$\Rightarrow \tan p - \cot q = 0$$~~

~~$$\Rightarrow \tan p - \frac{1}{\tan q} = 0$$~~

c) $\tan p = \cot q$

$$\Rightarrow \tan p = \tan \left(\frac{\pi}{2} - q\right)$$

The general solution is,

$$p = n\pi + \frac{\pi}{2} - q$$

$$p + q = n\pi + \frac{\pi}{2}$$

$$n(p+q) = n\pi + \frac{\pi}{2}$$

$$k n = \frac{n\pi + \frac{\pi}{2}}{p+q}$$

$$\therefore n = \frac{(2n+1)\pi}{p+q} \cdot \frac{1}{2}$$

8) Find the solution of

a) $\tan 2x = \tan x$ ($-\pi \leq x \leq \pi$)

$$\Rightarrow \tan 2x - \tan x = 0$$

$$\Rightarrow \frac{\tan 2x - \tan x}{1 + \tan 2x \cdot \tan x} = 0$$

$$\Rightarrow \tan (2x - x) = 0$$

$$\Rightarrow \tan x = 0$$

$$\Rightarrow \tan x = \tan 0, \tan -\pi, \tan \pi$$

$$\therefore x = 0, -\pi, \pi,$$

b) $\tan x - 3 \cot x = 2 \tan 3x$ for $0 \leq x \leq 2\pi$

$$\Rightarrow \tan x - \frac{3}{\tan x} = 2 \tan 3x$$

$$\Rightarrow \tan x - \frac{3}{\tan x} = 2 \left(\frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x} \right)$$

let $\tan x = t$,

$$\text{or } t - \frac{3}{t} = 2 \left(\frac{3t - t^3}{1 - 3t^2} \right)$$

$$\text{or } \frac{t^2 - 3}{2t} = \frac{3t - t^3}{1 - 3t^2}$$

$$\text{or } (t^2 - 3)(1 - 3t^2) = 6t^2 - 2t^4$$

$$\text{or } t^2 - 3 - 3t^4 + 9t^2 = 6t^2 - 2t^4$$

$$\text{or, } 3t^4 - 2t^4 + 6t^2 - 10t^2 + 3 = 0$$

$$\text{or, } t^4 - 4t^2 + 3 = 0$$

$$\text{or, } t^4 - 3t^2 - t^2 + 3 = 0$$

$$\text{or, } t^2(t^2 - 3) - 1(t^2 - 3) = 0$$

$$\text{or, } (t^2 - 1)(t^2 - 3) = 0$$

Either,

$$t^2 - 1 = 0$$

$$t^2 = 1$$

$$\tan^2 \theta = 1$$

$$\tan \theta = \pm 1$$

$$\therefore \tan \theta = \tan \frac{\pi}{4}, \tan \frac{3\pi}{4}, \tan \frac{5\pi}{4}, \tan \frac{7\pi}{4}$$

$$\therefore \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

or,

$$t^2 - 3 = 0$$

$$t^2 = 3$$

$$t = \sqrt{3}$$

$$\tan \theta = \sqrt{3}, -\sqrt{3}$$

$$\tan \theta = \tan \frac{\pi}{3}, \tan \frac{2\pi}{3}, \tan \frac{4\pi}{3}, \tan \frac{5\pi}{3}$$

$$\therefore \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\therefore \theta = \frac{\pi}{4}, \frac{\pi}{3}, \frac{3\pi}{4}, \frac{2\pi}{3}, \frac{5\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{4}, \frac{5\pi}{3},$$

9. Find the solution of the equation

$$\cot x + \cot y = 2 \rightarrow (i)$$

$$2 \sin x \cdot \sin y = 1 \rightarrow (ii)$$

From (i)

$$\frac{\cos x}{\sin x} + \frac{\cos y}{\sin y} = 2$$

$$\cos x \cdot \sin y + \sin x \cdot \cos y = 2 \sin x \cdot \sin y$$

$$\sin(x+y) = 1$$

$$\sin(x+y) = \sin \frac{\pi}{2}$$

$$x+y = \frac{\pi}{2} \rightarrow (iii)$$

Now,

eqn (ii) becomes

$$2 \cdot (\cos(x-y) - \cos(x+y)) = 1$$

$$\Rightarrow \cos(x-y) - \cos \frac{\pi}{2} = 1$$

$$\Rightarrow \cos(x-y) - 0 = 1$$

$$\Rightarrow \cos(x-y) = 1$$

$$\Rightarrow \cos(x-y) = \cos 0$$

$$x-y = 0$$

$$x = y$$

$\therefore$ From (iii)

$$x = \frac{\pi}{4} \text{ and } y = \frac{\pi}{4}$$